

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EC) Examination****May 2017****EC 201.01 / EC 201 Digital Electronics & Logic Design****Date: 23.05.2017, Tuesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the below questions.**

- a. Draw the Combinational logic diagram for Full Adder circuit with truth table and necessary functions. **[04]**
- b. Which two logic gates are known as Universal Gate? Justify answer with example. **[03]**

Q – 2 Answer the below questions.

- a. Write a short note on multiplexer (4x1) & de-multiplexer (1x4) with Logic Diagrams. **[07]**
- b. Design BCD to decimal decoder. **[07]**

OR**Q – 2 Answer the below questions.**

- a. Write a short note on 4 bit binary magnitude comparator with required diagram and equations. **[07]**
- b. Design BCD to Excess-3 code converter with Truth table and functions for 0 to 9 decimal numbers. **[07]**

Q – 3 Answer the below questions.

- a. Simplify following function using K-map **[07]**

$$F(w,x,y,z) = \sum(0,1,2,4,5,6,8,9,12,13,14)$$
- b. Simplify the following Boolean function by using the tabulation method: **[07]**

$$F(a,b,c,d) = \sum(0,1,2,8,10,11,14,15)$$

OR**Q – 3 Answer the below questions.**

- a. Simplify following function using K-map **[07]**

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$
- b. Determine the prime implicants of the function: **[07]**

$$F(w,x,y,z) = \sum(1,4,6,7,8,9,10,11,15)$$

SECTION – II**Q - 4 Answer the below questions.**

- a. Convert $(19.75)_{10}$ into binary. **[01]**
- b. Calculate: $(5531)_8 - (3261)_8 + (100)_{10} = (?)_{10}$ **[02]**

- c. Perform subtractions of BCD number using 10's complement: $24 - 68 = ?$ [02]
 d. Determine the base x : $(211)_x = (152)_8$ [02]

Q – 5 Answer the below questions.

- a. Convert S-R Flip flop to D Flip flop. [07]
 b. Implement Full Adder with 3x8 decoder. [07]

OR

Q – 5 Answer the below questions.

- a. Write down excitation tables for S-R, J-K, D and T flip flops with logic diagram. [07]
 b. Design a 4 bit synchronous UP / DOWN counter. [07]

Q – 6 Answer the below questions.

- a. Explain 4 bit binary parallel adder circuit with proper logic diagram. [07]
 b. Design following function using proper multiplexer (H, J and L must be select line.) [07]
 $F(H,J,K,L) = \sum(0,2,3,5,8,9,11,13,14,15)$

OR

Q – 6 Answer the below questions.

- a. Write a short note on switch-tail ring counter. [07]
 b. Design a combinational circuit using a ROM. The circuit accepts a 3 bit number and generates an output binary number equal to the square of the input number. [07]
